Yoav Gudes

AI/ML Engineering Co-op Student

University of Waterloo 2028

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Summary

- Waterloo Nanotechnology Engineering student (Nanoelectronics & Quantum Engineering) with 16 months of co-op.
- Built NLP classifiers and multi-step LLM pipelines end to end, from raw data and labeling to evaluation and production.
- Proficient in Python, SQL, Git and Azure, and in building with agentic AI tools including Claude, Codex and MCP.

Education

Nanotechnology Engineering: Nanoelectronics & Quantum Engineering Options | **University of Waterloo** **Sept 2023 – Apr 2028**

- **Relevant Coursework:** Calculus I–IV, Linear Algebra, Differential Equations, Numerical Methods, Probability & Statistics

AI Projects

- **Portfolio Website:** Built and deployed a portfolio site with Claude and Codex, using a Python build script and GitHub Pages.
- **MCP Integrations:** Connected Claude to Gmail, Google Calendar and Drive via MCP connectors to automate workflows.

Work Experience

Process Engineering Intern | WOLF Advanced Technology **Sept 2025 – Apr 2026**

- Assembled and diagnosed 50+ NVIDIA/AMD GPUs, validating bandwidth and thermal stability with CUDA and FurMark.
- Delegated multi-step research, report drafting and CAD documentation tasks to agentic AI tools (Codex, Claude).

AI/ML Engineering & Data Science Intern | TD Bank **Jan 2025 – Apr 2025**

- Developed an NLP sentiment classifier for client calls, building an Azure Databricks speech-to-text and parsing pipeline.
- Hand-labeled 10,000+ utterances; trained XGBoost on MPNet embeddings with class rebalancing and Optuna tuning.
- Benchmarked three model variants on precision, recall and F1, reaching up to 98.8% accuracy as an Azure AI alternative.

Data Engineering Intern | MedReddie **May 2024 – Aug 2024**

- Built a multi-step LLM pipeline (Gemini API) that found supplier websites, validated pages and extracted contacts.
- Wrote few-shot prompts with structured outputs and null fallbacks, chaining LLM calls with BeautifulSoup and Selenium.
- Collected data for 8,000+ suppliers and $50M+ in products, loading it into staging and production with SQL and Python.

Technical Skills

LLMs & Agentic AI

- *LLM Pipelines:* Multi-Step LLM API Calls in Python
- *Prompting:* Few-Shot, Structured Outputs, Fallbacks
- *Agentic Tools:* Claude, Codex, MCP Connectors

Machine Learning & Evaluation

- *ML Algorithms:* XGBoost, Random Forests
- *Embeddings:* MPNet Sentence Transformers, UMAP
- *Evaluation:* Optuna, GridSearchCV, F1, Error Analysis

Programming & Platforms

- *Languages:* Python, SQL, Java, MATLAB
- *Libraries:* Pandas, scikit-learn, XGBoost, Optuna
- *Platforms:* Azure Databricks, Azure AI, Jupyter, Git

Data Engineering

- *Scraping:* BeautifulSoup, Selenium, Requests
- *Data Quality:* Fuzzy Matching, Deduplication, Validation
- *Databases:* SQL, Staging & Production Environments

Leadership & Awards

President | Hillel Waterloo & Laurier **May 2025 – Aug 2026**

Awards

- **University of Waterloo:** Term Distinction (Dean's List), Excellent Academic Standing & Co-op Review, Jordan Banks Award.
- **TD Bank:** Achieve the Dream Canada Scholarship, Co-op Challenge Finalist